NATIONAL INSTITUTE OF TECHNOLGY PATNA

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Mid Semester Examination October 2023

B. Tech: Semester – 5th

Course Name: Software Engineering ***Course Code:* CS54116**

Maximum Time: 2 Hours **Max. Marks:30**

Instructions:

1. ***Answer all questions. Answer to sub questions must be given sequentially at one place.***
2. ***Question No.*5 has two parts, one for CSE section 1 and other for CSE section 2&3. Attempt only your concerned section part.**
3. Assume any suitable data, if necessary.
4. The Marks, CO (Course Outcome) related to questions are mentioned on the right-hand side margin.

Q. No.		Questions	Marks	CO
1.	a)	Reasons for project failure could be _____ and _____ .	(6)	CO2 /CO3
	b)	A fresher (new graduate) can be inducted as a System Architect. (True/False)		
	c)	How do you ensure 99% complete syndrome doesn't occur?		
	d)	Who are two stake holders of feasibility report?		
	e)	How is phase containment error avoided?		
	f)	Why Spiral model is called Meta model?		
2.	a)	Differentiate between Conventional Process Model and Evolutionary Process Model.	(3)	CO3
	b)	Consider the software project with a number (weighing factor) of input, output, inquiries, files and interfaces as 30(4), 25(5), 20(4), 10(10), 5(7) respectively. Assuming all complexity factors are average i.e., 3. i. What is the function point of the project? ii. Estimate the productivity if the effort required is 8 persons-month.	(3)	CO5
3.	a)	Discuss the major responsibilities and skills of a software project manager.	(3)	CO2
	b)	Consider a software project that was estimated to be 400KLOC. Calculate the effort and the development time for each of the three modes i.e. organic, semidetached and embedded using COCOMO.	(3)	CO5
4.	a)	List three common types of risks that a typical software project might suffer from. Explain how you can identify the risks that your project is susceptible to. Suppose you are the project manager of a large software development project, point out the main steps you would follow to manage risks in your software project.	(3)	CO2/ CO4
	b)	What are the different types of relationships that might exist among the classes in an object-oriented design? Draw a class diagram of the library management system using the UML syntax to represent the fact that it has three classes: librarian, member, and book. A member can be a student or staff.	(3)	CO1/ CO6
5.		**(For CSE Section -1 only)**		
	a)	A Software company is planning for a Zombie gaming software. The tasks of the software are given as follows,	(6)	CO2

Task	Time in Days	Predecessors
A. Robotic control module	5	---
B. Texture library	5	C
C. Texture editor	4	---
D. Character editor	6	A, G, I
E. Character animator	7	D
F. Artificial intelligence (for zombies)	7	---
G. Rendering engine	6	---
H. Humanoid base classes	3	---
I. Character classes	3	H
J. Zombie classes	3	H
K. Test environment	5	L
L. Test environment editor	6	C, G
M. Character library	9	B, E, I
N. Zombie library	15	B, J, O
O. Zombie editor	5	A, G, J
P. Zombie animator	6	O
Q. Character testing	4	K, M
R. Zombie testing	4	K, N

The management wants to optimize the cost by diverting the resources which could be delayed. So, assume that you are **Project Manager**, you have been asked by your Superior to plan a schedule of the activities properly and report the critical activities to them? Show all the details of scheduling that you would be reporting including which activities could be delayed and to what extent so that management can take a decision to optimize the cost.

(for CSE section II and III only)

c) Discuss the difference between object-oriented and function-oriented design. (3) CO-3

d) What are the relative advantages of using either the LOC or the function point metric to measure the size of a software product for software project planning. (3) CO-3

********** *ALL THE BEST* **********

GYB